

Practice Assignment on Classes and Objects

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Btech. CSE (Cyber) - 3rd Sem

Problem Statement: Real Estate Management System

Create a class named **Property** that includes the following members:

Private Data Members (Encapsulation):

- propertyID (integer) – Unique identification number.
- propertyType (string) – e.g., "Apartment", "House", "Condo".
- price (double) – Cost of the property.
- isAvailable (boolean) – Availability status (true if for sale, false if sold).

Public Member Functions:

- **void setDetails(int id, string type, double p)** – To initialize the properties' ID, type, and price. Newly set properties default to available (true).
- **void updatePrice(double newPrice)** – To update the property price to a new value.
- **void markAsSold()** – To change isAvailable to false.
- **void display()** – To print all property information cleanly to the console.

In your main() function, write code to execute the following sequential workflow:

- Instantiate **two** distinct objects of the Property class (e.g., p1 and p2).
- Set the details for both properties using your setter function.
- Display the initial state of both properties.
- Update the price of the first property.
- Change the status of the second property to sold.
- Display the updated state of both properties to prove the data has successfully modified.

Code :

```

#include <iostream>
#include <string>

using namespace std;

class Property
{
    //by default private :
    unsigned int propertyID;
    string propertyType;
    double price;
    bool isAvailable;

public :
    void setDetails(int id, string type, double p, bool avail=true)
    {
        propertyID=id;
        propertyType=type;
        price=p;
        isAvailable=avail;
    }
    void updatePrice(double newPrice)
    {
        price=newPrice;
    }
    void markAsSold()
    {
        isAvailable=false;
    }
    void display()
    {
        string avail = isAvailable?"Yes":"No";
        cout << endl << "\nProperty ID : " << propertyID
        << "\nProperty Type : " << propertyType
        << "\nPrice : " << price
        << "\nAvailablity : " << avail;
    }
};

int main()
{
    cout << endl << "~ ~ ~ ~ ~ ~ ~ ~ Real Estate Management System ~ ~ ~ ~ ~
~ ~ ~";

    Property p1,p2;
    int id1,id2;
    string type1,type2;
    double pr1,pr2;
    bool avail1,avail2;

    cout << "\n\nProperty 1\nProperty ID : ";
    cin >> id1;

```

```

    cout << "\nProperty Type : ";
    cin >> type1 ;
    cout << "\nProperty Price : ";
    cin >> pr1;
    p1.setDetails(id1,type1,pr1);

    label :
    cout << "\n\nProperty 2\nProperty ID : ";
    cin >> id2;
    while(id1==id2)
    {cout << "\nProperty ID can't be duplicated";
        goto label;
    }
    cout << "\nProperty Type : ";
    cin >> type2 ;
    cout << "\nProperty Price : ";
    cin >> pr2;
    p2.setDetails(id2,type2,pr2);

    cout << "\n\nInitial Details\n" ;
    p1.display();
    p2.display();

    cout << "\nEnter New Price of 1st Property : ";
    cin >> pr1;
    p1.updatePrice(pr2);

    cout << "\nPress Any Key and Enter to Change the Status of 2nd Property : ";
    cin >> type2;
    p2.markAsSold();

    cout << "\nUpdated Details\n";
    p1.display();
    p2.display();

    return 0;
}

```

Output :

~~~~~ Real Estate Management System ~~~~~

Property 1

Property ID : 101

Property Type : Condo

Property Price : 65000000

Property 2

Property ID : 101

Property ID can't be duplicated

Property 2

Property ID : 102

Property Type : Appartement

Property Price : 67000000

Initial Details

Property ID : 101

Property Type : Condo

Price : 6.5e+08

Availablity : Yes

Property ID : 102

Property Type : Appartement

Price : 6.7e+08

Availablity : Yes

Enter New Price of 1st Property : 3400000

Press Any Key and Enter to Change the Status of 2nd Property : f

Updated Details

Property ID : 101

Property Type : Condo

Price : 6.7e+08

Availablity : Yes

Property ID : 102

Property Type : Appartement

Price : 6.7e+08

Availablity : No